

§ 7. Arbitrary system $\mathfrak{S}_{n\rho}$, linear family $\mathfrak{L}_{n\rho}$, and integrality domain $\mathfrak{J}_{n\rho}$ of rational functions. Existence of the rational basis.

From the existence of the rational basis of the fields $\mathfrak{K}_{n\rho}$ we shall now obtain the rational basis for arbitrary systems of rational and integral rational functions. We arrange these systems in the order in which they successively arise from one another through the elementary operations of addition, multiplication, and division.

I. As always, let $\mathfrak{S}_{n\rho}$ denote an arbitrary system of rational (or integral rational) functions of n indeterminates and of algebraic rank ρ . The quantities from Ω are counted as belonging to the system.

II. The system $\mathfrak{L}_{n\rho}$ is called a *linear family* if it satisfies the conditions:

1. Together with $f(x)$, $\mathfrak{L}_{n\rho}$ always also contains $c \cdot f(x)$, where c is an arbitrary quantity from Ω ;
2. Together with $f(x)$ and $g(x)$, $\mathfrak{L}_{n\rho}$ always also contains $f(x) + g(x)$.

III. The linear family $\mathfrak{J}_{n\rho}$ is called an *integrality domain* under the further condition:

3. Together with $f(x)$ and $g(x)$, $\mathfrak{J}_{n\rho}$ always also contains $f(x) \cdot g(x)$.

IV. The integrality domain $\mathfrak{K}_{n\rho}$ is called a *field* under the further condition:

4. Together with $f(x)$ and $g(x)$ — where $g(x) \neq 0$ — $\mathfrak{K}_{n\rho}$ always also contains the quotient $f(x) : g(x)$.

Conditions 1 through 4 are those stated in § 1 for the field.¹

These domains are now to be derived successively from one another.

a) An arbitrary system $\mathfrak{S}_{n\rho}$ is enlarged to a linear family $\mathfrak{L}_{n\rho}$ by the requirement:

$\mathfrak{L}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented linearly with coefficients from Ω by a finite number of functions from $\mathfrak{S}_{n\rho}$:

$$h(x) = c_1 f_1(x) + c_2 f_2(x) + \cdots + c_k f_k(x),$$

where $f_1(x), \dots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$, and $c_1, \dots, c_k$ through all quantities from Ω .

For adding rational combinations of functions of the system preserves the algebraic rank of a system. $\mathfrak{L}_{n\rho}$ satisfies conditions 1 and 2, and is therefore a linear family. Moreover, every linear family containing $\mathfrak{S}_{n\rho}$ also contains $\mathfrak{L}_{n\rho}$ by 1 and 2; hence $\mathfrak{L}_{n\rho}$ is the smallest linear family containing $\mathfrak{S}_{n\rho}$.

b) Correspondingly, from a linear family $\mathfrak{L}_{n\rho}$ arises the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ by the requirement:

$\mathfrak{J}_{n\rho}$ contains, in addition to $\mathfrak{L}_{n\rho}$, all functions $h(x)$ and only those that can be represented as an *integral rational combination* — with coefficients from Ω — of a finite number of functions from $\mathfrak{L}_{n\rho}$, $f_1(x), \dots, f_k(x)$, where $f_1(x), \dots, f_k(x)$ run through all functions from $\mathfrak{L}_{n\rho}$.

c) Finally, from an integrality domain $\mathfrak{J}_{n\rho}$ arises the smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement: $\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{J}_{n\rho}$, all functions $h(x)$ and only those that can be represented as quotients of two functions from $\mathfrak{J}_{n\rho}$.

Combining the three extensions into one step gives:

Every system $\mathfrak{S}_{n\rho}$ can be enlarged to a smallest containing field $\mathfrak{K}_{n\rho}$ by the requirement:

$\mathfrak{K}_{n\rho}$ contains, in addition to $\mathfrak{S}_{n\rho}$, all functions $h(x)$ and only those that can be represented as rational combinations — with coefficients from Ω — of a finite number of functions from $\mathfrak{S}_{n\rho}$, $f_1(x), \dots, f_k(x)$, where $f_1(x), \dots, f_k(x)$ run through all functions from $\mathfrak{S}_{n\rho}$.

From the latter fact the existence of the rational basis for arbitrary domains follows immediately:

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{S}_{n\rho}$, and let a rational basis of $\mathfrak{K}_{n\rho}$ be given by

$$h_1(x), h_2(x), \dots, h_\sigma(x). \quad (1)$$

By the definition of $\mathfrak{K}_{n\rho}$, relations exist:

$$\begin{aligned} h_1(x) &= r_1(g_1(x), \dots, g_\lambda(x)); \dots; \\ h_\sigma(x) &= r_\sigma(g_\mu(x), \dots, g_\nu(x)), \end{aligned} \quad (2)$$

where $r_1, \dots, r_\sigma$ are rational functions of their arguments, and

$$g_1(x), \dots, g_\lambda(x), \dots, g_\mu(x), \dots, g_\nu(x) \quad (3)$$

all belong to $\mathfrak{S}_{n\rho}$. Because of the relations (2), (3), in addition to (1), also represents a rational basis of $\mathfrak{K}_{n\rho}$; because the functions (3) belong to $\mathfrak{S}_{n\rho}$, this is at the same time a rational basis of $\mathfrak{S}_{n\rho}$. We therefore have

¹ $f(x)$ and $g(x)$ may also be of degree zero, i.e. quantities from Ω .

Theorem IV: An arbitrary system $\mathfrak{S}_{n\rho}$ of rational (or integral rational) functions possesses a rational basis, consisting of a finite number of functions of the system, of algebraic rank ρ .

Now let $\mathfrak{L}_{n\rho}$ in particular be a linear family, and let

$$g_1(x), \dots, g_\rho(x), g_{\rho+1}(x), \dots, g_\nu(x) \quad (4)$$

be a rational basis of $\mathfrak{L}_{n\rho}$. Suppose, for instance, that

$$g_1(x), \dots, g_\rho(x)$$

are algebraically independent. If one then sets

$$g(x) = c_1 g_{\rho+1}(x) + c_2 g_{\rho+2}(x) + \dots + c_{\nu-\rho} g_\nu(x),$$

then the c_i can be chosen as numbers from Ω in such a way that

$$g_1(x), \dots, g_\rho(x), g(x) \quad (5)$$

becomes a rational basis of the smallest containing field $\mathfrak{K}_{n\rho}$. But since $\mathfrak{L}_{n\rho}$ is a linear family, $g(x)$ belongs to $\mathfrak{L}_{n\rho}$, and (5) represents a rational basis of $\mathfrak{L}_{n\rho}$; that is,

Theorem V: The rational basis of every linear family $\mathfrak{L}_{n\rho}$ of rational (or integral rational) functions — hence in particular the rational basis of every integrality domain $\mathfrak{I}_{n\rho}$ and every field $\mathfrak{K}_{n\rho}$ — can be chosen in particular so that it contains at most $\rho + 1$ functions (and at least ρ).

The bound on the number of functions in the rational basis found in § 4 for fields therefore rests on the property of the field of being a linear family, whereas the restriction to ρ functions in the case of the existence of the minimal basis uses all assumptions of the field.

It remains to note that, by § 3 (4), all characteristic properties of the systems and smallest containing systems are preserved under the mapping.

§ 8. Involution basis of the integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials.

In the following paragraphs we are concerned with systems $\mathfrak{S}_{n\rho}$ whose elements are integral rational functions (polynomials); first with integrality domains $\mathfrak{J}_{n\rho}$ consisting of polynomials. For these domains $\mathfrak{J}_{n\rho}$, the divisibility notions are first to be fixed.

Let $F(x), G(x)$ be two polynomials from $\mathfrak{J}_{n\rho}$, and let $L(x)$ be a common divisor of these polynomials, so that one has

$$F(x) = L(x) \cdot F_1(x), \quad G(x) = L(x) \cdot G_1(x).$$

$L(x)$ is called a *common divisor in $\mathfrak{J}_{n\rho}$* if and only if the three polynomials $L(x), F_1(x), G_1(x)$ belong to $\mathfrak{J}_{n\rho}$.

Two polynomials from $\mathfrak{J}_{n\rho}$ are called *relatively prime in $\mathfrak{J}_{n\rho}$* if they possess no common divisor in $\mathfrak{J}_{n\rho}$.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{J}_{n\rho}$. Then, from the definition, every function from $\mathfrak{K}_{n\rho}$ has a representation

$$f(x) = F_1(x) : F_2(x),$$

where $F_1(x), F_2(x)$ belong to $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$. Correspondingly, several functions from $\mathfrak{K}_{n\rho}$ can be brought to a common denominator in $\mathfrak{J}_{n\rho}$:

$$f_1(x) = \frac{F_1(x)}{F(x)}, \dots, f_k(x) = \frac{F_k(x)}{F(x)}.$$

$F(x)$ is called a *least common denominator in $\mathfrak{J}_{n\rho}$* if and only if the polynomials from $\mathfrak{J}_{n\rho}$,

$$F(x), F_1(x), \dots, F_k(x),$$

are relatively prime in $\mathfrak{J}_{n\rho}$.

Such a representation can be possible in different ways, since in $\mathfrak{J}_{n\rho}$ the laws of unique factorization into irreducible functions do not hold in general.

We now investigate the significance of the involution basis of $\mathfrak{K}_{n\rho}$ for $\mathfrak{J}_{n\rho}$, by transferring the notions of involution form and involution basis to integrality domains. Let

$$\Phi(z, u) = z^i + \sum' g_{\alpha\alpha_1\dots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \dots u_\rho^{\alpha_\rho}, \quad (\alpha + \alpha_1 + \dots + \alpha_\rho = i)$$

be an involution form of $\mathfrak{K}_{n\rho}$, and let $G(x)$ be a least common denominator in $\mathfrak{J}_{n\rho}$ of the coefficients of $\Phi(z, u)$. By multiplication with $G(x)$, $\Phi(z, u)$ passes into a form $\Psi(z, u)$ whose coefficients are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$:

$$G(x) \cdot \Phi(z, u) = \Psi(z, u) = G(x) z^i + \sum' G_{\alpha\alpha_1\dots\alpha_\rho}(x) z^\alpha u_1^{\alpha_1} \dots u_\rho^{\alpha_\rho}, \quad (\alpha + \alpha_1 + \dots + \alpha_\rho = i).$$

Every form $\Psi(z, u)$ arising from an involution form $\Phi(z, u)$ of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$ in such a way that the coefficients of $\Psi(z, u)$ are polynomials from $\mathfrak{J}_{n\rho}$ and are relatively prime in $\mathfrak{J}_{n\rho}$, shall be called an *involution form* of $\mathfrak{J}_{n\rho}$; and the coefficients of $\Psi(z, u)$ shall be called a corresponding *involution basis*.

It is first clear from the meaning of the involution basis of $\mathfrak{K}_{n\rho}$ that every involution basis of $\mathfrak{J}_{n\rho}$ is at the same time a rational basis. For the further investigation we consider a mapping domain $\mathfrak{J}_{\rho\rho}^*$ for whose smallest containing field $\mathfrak{K}_{\rho\rho}^*$ the irreducible equation with zero

$$z = u_1 x_1 + \dots + u_\rho x_\rho$$

is given by $\Phi^*(z, u) = 0$.

According to § 5, the coefficients of

$$\Phi^*(z, u) = z^i + \sum' g_{\alpha\alpha_1\dots\alpha_\rho}^*(x) z^\alpha u_1^{\alpha_1} \dots u_\rho^{\alpha_\rho}, \quad (\alpha + \alpha_1 + \dots + \alpha_\rho = i)$$

represent the elementary symmetric functions of the rows of quantities conjugate with respect to $\mathfrak{K}_{\rho\rho}^*$,

$$x, \quad x^{(1)}, \dots, x^{(i-1)}.$$

Let $F^*(x)$ be any polynomial from $\mathfrak{K}_{\rho\rho}^*$. Since

$$F^*(x) = \frac{1}{i} \left(F^*(x) + F^*(x^{(1)}) + \dots + F^*(x^{(i-1)}) \right),$$

it follows that $F^*(x)$ is an integral symmetric function of these rows of quantities; consequently it can be expressed as an integral rational combination of the elementary symmetric functions, that is, of the functions

$$g_{\alpha\alpha_1\ldots\alpha_\rho}^*(x).$$

If now

$$G^*(x) \cdot \Phi^*(z, u) = \Psi^*(z, u) = G^*(x)z^i + \sum' G_{\alpha\alpha_1\ldots\alpha_\rho}^*(x)z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}, \quad (\alpha + \alpha_1 + \cdots + \alpha_\rho = i)$$

is an involution form of $\mathfrak{J}_{\rho\rho}^*$, then every polynomial from $\mathfrak{K}_{\rho\rho}^*$, and in particular every polynomial from $\mathfrak{J}_{\rho\rho}^*$, is an integral rational combination of the quotients

$$G_{\alpha\alpha_1\ldots\alpha_\rho}^*(x) : G^*(x).$$

Taking the transfer principle into account, this gives

Theorem VI: To every integrality domain of polynomials $\mathfrak{J}_{n\rho}$ there exists at least one distinguished rational basis, the involution basis, such that in the rational representation of all polynomials from $\mathfrak{J}_{n\rho}$ by the functions of the involution basis only a power of one fixed function (the highest coefficient of the corresponding involution form) occurs in the denominator.

§ 9. Relatively integral domains $\mathfrak{G}_{n\rho}$ consisting of polynomials.

In what follows we consider special integrality domains consisting of polynomials, which form an intermediate stage between integrality domains and fields.

An integrality domain of polynomials $\mathfrak{G}_{n\rho}$ is called a “relatively integral domain” under the condition:

4a) $\mathfrak{G}_{n\rho}$ contains, along with $F(x)$ and $G(x)$ — where $G(x) \neq 0$ — the quotient $F(x) : G(x)$ always and only when this quotient is integral in the indeterminates, that is, is a polynomial.

First the identity of the relatively integral domains with the totality of the polynomials of a field $\mathfrak{K}_{n\sigma}$ ($\sigma \geq \rho$) of rational functions is to be proved.

Let $\mathfrak{K}_{n\rho}$ be the smallest field containing $\mathfrak{G}_{n\rho}$. For every function $f(x)$ from $\mathfrak{K}_{n\rho}$ there is a representation as quotient of two polynomials from $\mathfrak{G}_{n\rho}$:

$$f(x) = F_1(x) : F_2(x),$$

where polynomials of degree zero are also allowed. As soon as $f(x)$ becomes a polynomial, it belongs to $\mathfrak{G}_{n\rho}$ by condition 4a); and since $\mathfrak{G}_{n\rho}$ can contain no further polynomials, *every relatively integral domain $\mathfrak{G}_{n\rho}$ is identical with the totality of the polynomials of the smallest containing field.*

Conversely, let $\mathfrak{K}_{n\sigma}$ be any field of rational functions (therefore not necessarily the smallest containing field of a given system), and let $\mathfrak{J}$ be the totality of the polynomials contained in $\mathfrak{K}_{n\sigma}$. $\mathfrak{J}$ satisfies the conditions 1, 2, 3 of an integrality domain and condition 4a), and so is a relatively integral domain; that is, *the totality of the polynomials contained in a field $\mathfrak{K}_{n\sigma}$ forms a relatively integral domain $\mathfrak{G}_{n\rho}$ ($\rho \leq \sigma$).*²

We further show the identity of the relatively integral domains with Hilbert’s “integrality domains of relatively integral functions.”³

Let

$$G_1(x), \ldots, G_k(x) \tag{1}$$

be a rational basis of $\mathfrak{G}_{n\rho}$. By conditions 1, 2, 3, 4a, every rational combination of the polynomials (1) which is integral in the indeterminates, that is, becomes a polynomial, belongs to $\mathfrak{G}_{n\rho}$. Conversely, every polynomial from $\mathfrak{G}_{n\rho}$ can, by the notion of rational basis, be rationally expressed through the polynomials (1); hence $\mathfrak{G}_{n\rho}$ is identical with the totality of those rational combinations of the polynomials (1) which are integral in the indeterminates, that is, are polynomials. Thus we have Hilbert’s definition starting from the special rational basis, whereas our definition uses only abstract properties of the domain.

For relatively integral domains, the definition of a common divisor given in § 8 for general integrality domains simplifies:

Let $F(x), G(x)$ be two polynomials from $\mathfrak{G}_{n\rho}$ and let $L(x)$ be a common divisor of these polynomials. Then $L(x)$ is called a *common divisor in $\mathfrak{G}_{n\rho}$* if and only if $L(x)$ belongs to $\mathfrak{G}_{n\rho}$.

²It may also be $\rho = 0$; that is, the totality of the polynomials from $\mathfrak{K}_{n\sigma}$ consists of elements of the coefficient domain Ω .

³D. Hilbert: Mathematische Probleme. Lecture Paris 1900. Problem 14 (Göttinger Nachrichten 1900).

For then the membership of the quotients $F(x) : L(x)$ and $G(x) : L(x)$ follows from condition 4a).

Another essential property of relatively integral domains is that the notion of being “algebraically integral with respect to $\mathfrak{G}_{n\rho}$ ” can be defined by means of any equation exactly as is known for algebraic number fields or for the field $\Omega(x_1, \dots, x_n)$.

Definition V: A quantity ξ is called algebraically integral with respect to $\mathfrak{G}_{n\rho}$ — or also relatively algebraically integral — if it satisfies some equation whose highest coefficient is the unit and whose remaining polynomials are from $\mathfrak{G}_{n\rho}$.

Suppose, namely, that the relatively algebraically integral quantity ξ satisfies, for example, the equation

$$L(z) = z^\lambda + G_1(x)z^{\lambda-1} + \dots + G_\lambda(x) = 0.$$

$L(z)$ is divisible by the integral function irreducible with respect to the smallest containing field $\mathfrak{K}_{n\rho}$,

$$M(z) = z^\mu + H_1(x)z^{\mu-1} + \dots + H_\mu(x).$$

But from this divisibility it follows by the known extension of Gauss’s theorem⁴ that the rational functions $H_1(x), \dots, H_\mu(x)$ are polynomials from $\mathfrak{K}_{n\rho}$, and therefore belong to $\mathfrak{G}_{n\rho}$. Thus the definition of the relatively algebraically integral quantity ξ is obtained from the irreducible equation. In particular, every polynomial $F(x)$ from $\mathfrak{G}_{n\rho}$ is also algebraically integral with respect to $\mathfrak{G}_{n\rho}$, since it satisfies the equation $z - F(x) = 0$.

It should further be mentioned that, for an integrality domain $\mathfrak{J}_{n\rho}$ of polynomials, analogously to the smallest containing field, the smallest containing relatively integral domain $\mathfrak{G}_{n\rho}$ is also defined by the requirement:

$\mathfrak{G}_{n\rho}$ contains, along with $\mathfrak{J}_{n\rho}$, all polynomials $H(x)$, and only those, which can be represented as quotients of two polynomials from $\mathfrak{J}_{n\rho}$.

From this definition it follows further: From every involution form and involution basis of $\mathfrak{J}_{n\rho}$ a corresponding one of $\mathfrak{G}_{n\rho}$ arises by splitting off, at most, a common divisor in $\mathfrak{G}_{n\rho}$ of all polynomials of the basis.

For $\mathfrak{J}_{n\rho}$ and $\mathfrak{G}_{n\rho}$ possess the same smallest containing field $\mathfrak{K}_{n\rho}$; and every involution form of $\mathfrak{J}_{n\rho}$ arises from an involution form of $\mathfrak{K}_{n\rho}$ by multiplication with a polynomial from $\mathfrak{J}_{n\rho}$, and so has coefficients from $\mathfrak{G}_{n\rho}$, which however may have a common divisor in $\mathfrak{G}_{n\rho}$.⁵

Finally it should be noted that the *mapping domain* of a relatively integral domain $\mathfrak{G}_{n\rho}$, which by § 7 is again an integrality domain $\mathfrak{J}_{\rho\rho}^*$ of polynomials, need not itself be a relatively integral domain. For if $F(x)$ and $G(x)$ are two polynomials from $\mathfrak{G}_{n\rho}$ whose quotient is not a polynomial and consequently does not belong to $\mathfrak{G}_{n\rho}$, then the quotient $F^*(x) : G^*(x)$ also does not belong to $\mathfrak{J}_{\rho\rho}^*$. But this quotient, arising by specialization of some indeterminates, may very well be a polynomial, and for $\mathfrak{J}_{\rho\rho}^*$ condition 4a) is then not fulfilled.⁶

§ 10. Relatively integral domains of the first kind and their integrality basis.

The notion introduced in § 9 (Definition V), “relatively algebraically integral,” makes it possible to give a class of relatively integral domains which are finite integrality domains, and thereby to answer positively, at least for this class, a question raised by D. Hilbert (Math. Problems, Problem 14). We give the following

Definition VI: A finite number of polynomials from $\mathfrak{G}_{n\rho}$ is called an *integrality basis* if every polynomial from $\mathfrak{G}_{n\rho}$ can be represented as an integral rational combination of this finite number, with coefficients from

⁴Compare, for instance, J. König: Einleitung in die allgemeine Theorie der algebraischen Größen (Leipzig, B. G. Teubner, 1903), Kap. IX, § 2, or Weber (small edition), § 20.

⁵For example, for the integrality domain $\mathfrak{J}_{22}$ consisting of all integral rational combinations of x_1^2 and x_1x_2 , the following is obtained as an involution form:

$$\Psi(z, u) = x_1^2z^2 - x_1^4u_1^2 - 2x_1^3x_2u_1u_2 - x_1^2x_2^2u_2^2.$$

Since x_1^2 belongs to $\mathfrak{J}_{22}$, and therefore all the more to the smallest containing relatively integral domain $\mathfrak{G}_{22}$, and consequently is a common divisor in $\mathfrak{G}_{22}$, the following results as an involution form of $\mathfrak{G}_{22}$:

$$\Psi(z, u) = z^2 - x_1^2u_1^2 - 2x_1x_2u_1u_2 - x_2^2u_2^2.$$

⁶Let $\mathfrak{G}_{22}$ consist, for example, of all polynomials which are rational combinations of $F = x_1^2x_2$ and $G = x_2^2(1 + x_3)$. The admissible specialization $x_3 = 0$ gives $F^* : G^* = x_2$, whereas $F : G$ is not a polynomial. $\mathfrak{J}_{22}^*$ is therefore not a relatively integral domain.

Ω . An integrality domain of polynomials which possesses an integrality basis is called a finite integrality domain.

The class of relatively integral domains to be considered, for which the involution basis will prove to be an integrality basis, we designate as relatively integral domains of the first kind according to the following

Definition VII: A relatively integral domain $\mathfrak{G}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{G}_{\rho\rho}^*$ such that every arbitrary polynomial in $x_1, \dots, x_\rho$ (with coefficients from Ω) is algebraically integral over $\mathfrak{G}_{\rho\rho}^*$.

According to this definition, the indeterminates $x_1, \dots, x_\rho$, and consequently also the linear form $u(x) = u_1x_1 + \dots + u_\rho x_\rho$, are algebraically integral over $\mathfrak{G}_{\rho\rho}^*$. The irreducible integral function with zero $z = u(x)$ — the involution form of the smallest containing field $\mathfrak{K}_{\rho\rho}^*$ — therefore has the unit as highest coefficient and, as its further coefficients, polynomials from $\mathfrak{G}_{\rho\rho}^*$; it is therefore at the same time an involution form of $\mathfrak{G}_{\rho\rho}^*$, and in $\mathfrak{G}_{\rho\rho}^*$ no further involution form can exist. By the transfer principle, therefore, $\mathfrak{G}_{n\rho}$ also possesses an involution form whose highest coefficient is the unit; and since, by Theorem VI, in the rational representation of all polynomials from $\mathfrak{G}_{n\rho}$ by the corresponding involution basis only this highest coefficient enters into the denominator, the involution basis becomes an integrality basis. Thus:

Theorem VII. Every relatively integral domain of the first kind is a finite integrality domain; the integrality basis is given by the involution basis obtained from the characterizing mapping domain. In particular, the integrality basis of relatively integral domains of the first kind $\mathfrak{G}_{\rho\rho}$ is given by the uniquely defined involution basis of $\mathfrak{G}_{\rho\rho}$.

We add a criterion for relatively integral domains of the first kind. Let $\mathfrak{G}_{n\rho}$ be of the first kind and let an integrality basis of $\mathfrak{G}_{n\rho}$ be given by

$$F_1(x), F_2(x), \dots, F_k(x). \quad (1)$$

Then, by a Hilbert auxiliary theorem,⁷ there exist ρ algebraically independent polynomials from $\mathfrak{G}_{n\rho}$,

$$G_1(x), G_2(x), \dots, G_\rho(x), \quad (2)$$

such that the polynomials (1), and hence every polynomial from $\mathfrak{G}_{n\rho}$, are algebraically integral over the polynomials (2). The same holds for every polynomial of the mapping domain $\mathfrak{G}_{\rho\rho}^*$ with respect to the corresponding polynomials

$$G_1^*(x), G_2^*(x), \dots, G_\rho^*(x). \quad (3)$$

By Definition VII, therefore, every arbitrary polynomial in $x_1, \dots, x_\rho$ is also algebraically integral over the polynomials (3).

Conversely, suppose a mapping domain $\mathfrak{J}_{\rho\rho}^*$ of an arbitrary relatively integral domain $\mathfrak{G}_{n\rho}$ contains ρ polynomials (3) over which every polynomial in $x_1, \dots, x_\rho$ is algebraically integral. Then we show that $\mathfrak{J}_{\rho\rho}^*$ thereby becomes a relatively integral domain $\mathfrak{G}_{\rho\rho}^*$, and further that $\mathfrak{G}_{\rho\rho}^*$, and consequently $\mathfrak{G}_{n\rho}$, are of the first kind.

Indeed, let $F^*(x)$ be a polynomial from the smallest field $\mathfrak{K}_{\rho\rho}^*$ containing $\mathfrak{J}_{\rho\rho}^*$. By assumption it is algebraically integral over the polynomials (3). Since the corresponding function from $\mathfrak{K}_{n\rho}$ is then algebraically integral over ρ polynomials from $\mathfrak{G}_{n\rho}$, it is a polynomial $F(x)$ and hence belongs to $\mathfrak{G}_{n\rho}$; therefore $F^*(x)$ belongs to $\mathfrak{J}_{\rho\rho}^*$. The mapping domain $\mathfrak{J}_{\rho\rho}^*$ therefore contains all polynomials of $\mathfrak{K}_{\rho\rho}^*$ and is a relatively integral domain $\mathfrak{G}_{\rho\rho}^*$. For $\mathfrak{G}_{\rho\rho}^*$, and hence also for $\mathfrak{G}_{n\rho}$, it follows immediately from Definitions V and VII and from the assumption that they are of the first kind; that is:

The relatively integral domains of the first kind $\mathfrak{G}_{n\rho}$ are characterized by the fact that in a mapping domain $\mathfrak{J}_{\rho\rho}^$ there exist ρ polynomials over which every arbitrary polynomial in $x_1, \dots, x_\rho$ is algebraically integral. From this assumption the mapping domain itself results as a relatively integral domain.*⁸

§ 11. Auxiliary theorem on algebraically integral dependence.

We now give a criterion for every arbitrary polynomial in $x_1, \dots, x_\rho$ to be algebraically integral over ρ polynomials in $x_1, \dots, x_\rho$,

$$G_1^*(x), \dots, G_\rho^*(x). \quad (1)$$

⁷D. Hilbert: Über die vollen Invariantensysteme, Math. Ann. 42 (1893), § 1. The auxiliary theorem stated there only for homogeneous polynomials holds likewise for inhomogeneous ones.

⁸For example, in the domain $\mathfrak{G}_{22}$ mentioned in the closing remark to § 9, the admissible specialization $x_1 = 1$ gives the two polynomials $F^* = x_2$, $G^* = 1 + x_3$, over which every arbitrary polynomial in x_2, x_3 is algebraically integral. $\mathfrak{G}_{22}$ is therefore of the first kind, indeed with $F(x)$ and $G(x)$ as integrality basis.

If the polynomials (1) are, in particular, homogeneous forms, then for every special system of values which makes (1) vanish, all algebraically integrally dependent forms vanish at the same time; hence in particular also $x_1, \dots, x_\rho$.

The forms (1), therefore, cannot vanish for any system of values other than

$$x_1 = 0, \dots, x_\rho = 0;$$

their resultant must be different from zero.

Conversely, this condition is also sufficient and admits an extension to inhomogeneous polynomials according to the following

Auxiliary theorem: A sufficient condition that every arbitrary polynomial in $x_1, \dots, x_\rho$ be algebraically integral over ρ polynomials in $x_1, \dots, x_\rho$

$$G_1^*(x), \dots, G_\rho^*(x)$$

is the non-vanishing of the resultant, taken with respect to $x_1, \dots, x_\rho$, of the terms of highest dimension

$$H_1^*(x), \dots, H_\rho^*(x)$$

of the polynomials (1):

$$R_0 = \begin{bmatrix} H_1^* & \cdots & H_\rho^* \\ x_1 & \cdots & x_\rho \end{bmatrix} \neq 0. \quad (2)$$

*For homogeneous forms (1) the condition (2) is necessary.*⁹

For the proof we form the polynomials

$$\begin{aligned} \bar{G}_1^*(x) &= G_1^*(x) - \lambda_1, \dots, \bar{G}_\rho^*(x) = G_\rho^*(x) - \lambda_\rho, \\ \bar{\Phi}^*(x) &= \Phi^*(x) - \mu, \end{aligned} \quad (3)$$

where $\lambda_1, \dots, \lambda_\rho, \mu$ denote indeterminates and $\Phi^*(x)$ denotes an arbitrary polynomial in $x_1, \dots, x_\rho$. Let the resultant of the polynomials (3) with respect to $x_1, \dots, x_\rho$,¹⁰ be given by

$$R(\lambda, \mu) = \begin{bmatrix} \bar{G}_1^* & \cdots & \bar{G}_\rho^* & \bar{\Phi}^* \\ x_1 & \cdots & x_\rho & 1 \end{bmatrix}; \quad (4)$$

it is an integral rational function of $\lambda_1, \dots, \lambda_\rho, \mu$. According to F. Mertens, in the expansion of (4) by powers of μ , the highest coefficient can be explicitly specified;¹¹ one obtains

$$(-1)^\tau R(\lambda, \mu) = R_0^\sigma \mu^\tau + A_1(\lambda) \mu^{\tau-1} + \cdots + A_\tau(\lambda), \quad (5)$$

where $A_1(\lambda), \dots, A_\tau(\lambda)$ denote integral integer-valued functions of $\lambda_1, \dots, \lambda_\rho$ and the coefficients of the polynomials (3). This expansion first shows that under the assumption (2), $R_0 \neq 0$, also $R(\lambda, \mu)$ cannot vanish identically. The identity in $x_1, \dots, x_\rho, \lambda_1, \dots, \lambda_\rho, \mu$,

$$R(\lambda, \mu) \equiv 0 \pmod{\bar{G}_1^*(x), \dots, \bar{G}_\rho^*(x), \bar{\Phi}^*(x)},$$

passes, by the substitution

$$\lambda_i = G_i^*(x), \quad \mu = \Phi^*(x),$$

into

$$\begin{aligned} R(G_i^*(x), \Phi^*(x)) &= R_0^\sigma \Phi^{*\tau}(x) + A_1(G_i^*(x)) \Phi^{*\tau-1}(x) + \cdots \\ &\quad + A_\tau(G_i^*(x)) = 0, \end{aligned} \quad (6)$$

and this proves the auxiliary theorem, since R_0 is, by assumption, a number from Ω .

⁹For resultant theory, compare F. Mertens, Zur Theorie der Elimination (I. and II. Teil), Sitzungsber. d. k. Ak. d. Wiss. Wien, Math.-naturw. Kl. Abt. IIa, Bd. 108 (1899). Partly reproduced in J. König, Theorie d. algebr. Größen, Kap. VI.

¹⁰That is, the homogeneous resultant with respect to $x_1, \dots, x_\rho, x_0$, if the polynomials (3) are homogenized by means of an additional indeterminate x_0 so that the degree in the x is preserved.

¹¹A. a. O. § 3 (proof of Theorem I), and explicitly § 6 (determination of the totality of all terms of the resultant R which contain a given power product $a_{\lambda\lambda}^{p_\lambda} a_{\mu\mu}^{p_\mu} \cdots a_{\sigma\sigma}^{p_\sigma}$ as factor). In our case $p_\lambda = 0, p_\mu = 0, \dots, p_\rho = \tau, a_{\sigma\sigma} = (-\mu)$ is set. Reproduced in König, Kap. VI, § 9.

§ 12. The integrality basis of regular systems $\mathfrak{S}_{n\rho}$ consisting of polynomials.

The auxiliary theorem of § 11, together with the conclusion of § 10, immediately yields the fact:

If there exist, in a mapping domain $\mathfrak{J}_{\rho\rho}^$ of $\mathfrak{S}_{n\rho}$, ρ polynomials such that the homogeneous resultant of their terms of highest dimension does not vanish identically — $R_0 \neq 0$ —, then $\mathfrak{S}_{n\rho}$ is of the first kind and consequently is a finite integrality domain, with the involution basis as integrality basis.*

The condition $R_0 \neq 0$ now permits, by means of the auxiliary theorem, a second existence proof for the integrality basis, essentially due to D. Hilbert;¹² this proof, to be sure, does not make the involution basis recognizable as such, but it holds uniformly for all systems $\mathfrak{S}_{n\rho}$ with the indicated property.

Indeed, let the ρ polynomials from $\mathfrak{J}_{\rho\rho}^*$ for which $R_0 \neq 0$, and which consequently are algebraically independent, be given by

$$G_1^*(x), \dots, G_\rho^*(x). \quad (1)$$

By the auxiliary theorem, every arbitrary polynomial in $x_1, \dots, x_\rho$, hence in particular every polynomial of the smallest field $\mathfrak{K}_{\rho\rho}^*$ containing $\mathfrak{J}_{\rho\rho}^*$, is algebraically integral over the polynomials (1). If the corresponding polynomials in $\mathfrak{S}_{n\rho}$ are therefore

$$G_1(x), \dots, G_\rho(x), \quad (2)$$

then, by the transfer principle, every polynomial of the smallest field $\mathfrak{K}_{n\rho}$ containing $\mathfrak{S}_{n\rho}$, and hence in particular every polynomial from $\mathfrak{S}_{n\rho}$, is algebraically integral over the polynomials (2). Conversely, every function from $\mathfrak{K}_{n\rho}$ which is algebraically integral over the polynomials (2) is also a polynomial. Thus, if one regards $\mathfrak{K}_{n\rho}$ (according to § 4) as an algebraic field over $\Omega(G_1, \dots, G_\rho)$, the totality of the polynomials from $\mathfrak{K}_{n\rho}$ — the relatively integral domain $\mathfrak{S}_{n\rho}$ — is identical with the algebraically integral quantities of this field. Let $G(x)$ be a polynomial from $\mathfrak{K}_{n\rho}$ which completes (2) to a rational basis, and let $D(x)$ be the discriminant of the irreducible equation of degree k which $G(x)$ satisfies with respect to $\Omega(G_1, \dots, G_\rho)$. Then every polynomial from $\mathfrak{K}_{n\rho}$, and in particular every polynomial $F(x)$ from $\mathfrak{S}_{n\rho}$, as an algebraically integral quantity, admits the representation

$$F(x) = \frac{\Gamma_1(G_i)G(x)^{k-1} + \Gamma_2(G_i)G(x)^{k-2} + \dots + \Gamma_k(G_i)}{D(x)}. \quad (3)$$

Let $F(x)$ now run through all polynomials from $\mathfrak{S}_{n\rho}$. Then the application of Hilbert's Theorem I on the module basis (Math. Ann. 36) to the system formed from the corresponding functions

$$v_1\Gamma_1(G_i) + v_2\Gamma_2(G_i) + \dots + v_k\Gamma_k(G_i)$$

shows the existence of a finite number of polynomials from $\mathfrak{S}_{n\rho}$,

$$F_1(x), F_2(x), \dots, F_\lambda(x), \quad (4)$$

such that every polynomial from $\mathfrak{S}_{n\rho}$ admits a representation

$$F(x) = A_1(G_i)F_1(x) + A_2(G_i)F_2(x) + \dots + A_\lambda(G_i)F_\lambda(x), \quad (5)$$

where $A_1, \dots, A_\lambda$ denote polynomials in the $G_i(x)$. The polynomials (2) and (4) therefore form an integrality basis of $\mathfrak{S}_{n\rho}$; that is:

Theorem VIII: If there exist, in a mapping system $\mathfrak{J}_{\rho\rho}^$ of the system of polynomials $\mathfrak{S}_{n\rho}$, ρ polynomials such that the resultant of the terms of highest dimension does not vanish identically — $R_0 \neq 0$ —, then $\mathfrak{S}_{n\rho}$ possesses an integrality basis.*

This theorem allows a slight generalization. It is enough that the ρ polynomials (1) for which $R_0 \neq 0$ belong to the integrality domain $\mathfrak{J}_{\rho\rho}^*$ which is the smallest containing domain of $\mathfrak{S}_{\rho\rho}^*$. In fact, the arguments leading to Theorem VIII show that the representation (5) still holds then. But by the definition, every polynomial $L(x)$ of the smallest containing integrality domain $\mathfrak{J}_{n\rho}$ can be represented integrally and rationally by a finite number — depending on $L(x)$ — of polynomials from $\mathfrak{S}_{n\rho}$; hence there exist σ polynomials from $\mathfrak{S}_{n\rho}$,

$$K_1(x), K_2(x), \dots, K_\sigma(x), \quad (6)$$

¹²Über die vollen Invariantensysteme, § 2. In Hilbert's case, under the assumption of the otherwise obtained existence of the integrality basis of the invariant field, the point is only an interpretation of this basis through the Kronecker fundamental system of the algebraically integral quantities of a field.

such that the polynomials $G_i(x)$ become integral rational combinations of (6). Thus (6) and (4) give an integrality basis. Introducing the following terminology:

Definition VIII: A system $\mathfrak{S}_{n\rho}$ of polynomials is called regular if in a mapping domain $\mathfrak{J}_{\rho\rho}^$ of the smallest containing integrality domain there exist ρ polynomials such that the resultant of their terms of highest dimension does not vanish identically — $R_0 \neq 0$ —*

we therefore have:

*Theorem IX: Every regular system $\mathfrak{S}_{n\rho}$ of polynomials possesses an integrality basis.*¹³

We further give, analogously to § 5 for the involution, a geometric interpretation of regular systems. For this purpose we consider, as there in (7), the system of equations

$$L^*(x) - L^*(\xi) = 0, \quad (7)$$

where $L^*(x)$ runs through all polynomials from $\mathfrak{J}_{\rho\rho}^*$. Homogenizing by means of x_0, ξ_0 , let (7) pass into the system of equations between homogeneous forms

$$\xi_0^m M^*(x) - x_0^m M^*(\xi) = 0, \quad (8)$$

where, if $H^*(x)$ denotes the terms of highest dimension of $L^*(x)$, the relation

$$H^*(x) = M^*(x) \pmod{x_0} \quad (9)$$

holds.

As *fundamental points* of $\mathfrak{J}_{\rho\rho}^*$ (compare the note to § 5), we designate the solutions of (8) that are independent of x — thus different from the rows conjugate to x . Fundamental points are therefore the systems of values different from $\xi_1 = 0, \dots, \xi_\rho = 0$ for which simultaneously

$$\xi_0^m = 0, \quad M^*(\xi) = 0$$

holds, or, by (9),

$$\xi_0 = 0, \quad H^*(\xi) = 0.$$

If now there exist ρ forms $H^*(\xi)$ for which $R_0 \neq 0$, hence with no common system of solutions, then $\mathfrak{J}_{\rho\rho}^*$ can possess no fundamental point. If, in particular, $\mathfrak{S}_{\rho\rho}^*$ consists of homogeneous forms, then for $R_0 \neq 0$ also $\mathfrak{S}_{\rho\rho}^*$ can possess no fundamental point, since such a point would at the same time be a fundamental point of $\mathfrak{J}_{\rho\rho}^*$. But the converse also holds. For this we consider the system $\mathfrak{J}(H^*)$ formed from the totality of the forms $H^*(x)$; this is again an integrality domain.

From Hilbert's Theorem I on the module basis and Hilbert's auxiliary theorem (compare the citation in § 10), there follows the existence of ρ forms from $\mathfrak{J}(H^*)$ whose vanishing has the vanishing of all forms from $\mathfrak{J}(H^*)$ as consequence. If now $\mathfrak{J}_{\rho\rho}^*$ possesses no fundamental point, these ρ forms cannot vanish simultaneously; their resultant $R_0 \neq 0$.

Thus:

The regular systems $\mathfrak{S}_{n\rho}$ of polynomials are characterized by the fact that a mapping domain $\mathfrak{J}_{\rho\rho}^$ of the smallest containing integrality domain possesses no fundamental point. For regular systems $\mathfrak{S}_{\rho\rho}^*$ of homogeneous forms, the system itself already possesses no fundamental point.*¹⁴

¹³That there are non-regular systems without integrality basis was shown by Hilbert with an example (Gött. Nachr. 1891, p. 233). The following is even simpler:

$$\mathfrak{S}_{22} = x_1, x_1 x_2, x_1 x_2^2, \dots, x_1 x_2^\nu, \dots \quad \text{in inf.}$$

The existence of the integrality basis here would be equivalent to the existence of a positive integer ν such that the identities

$$x_1 x_2^{\nu+\sigma} = \sum C \cdot x_1^{i_0} (x_1 x_2)^{i_1} (x_1 x_2^2)^{i_2} \dots (x_1 x_2^\nu)^{i_\nu} \quad (\sigma = 1, 2, \dots)$$

hold. Comparing the dimensions in x_1, x_2 already for $\sigma = 1$ leads to the two conditional equations for the exponents

$$1 = i_0 + i_1 + \dots + i_\nu, \quad \nu + 1 = i_1 + 2i_2 + \dots + \nu i_\nu,$$

which plainly have no solution in non-negative integers. Thus $\mathfrak{S}_{22}$ possesses no integrality basis.

¹⁴The non-regular system $\mathfrak{S}_{22}$ given in the preceding note possesses the fundamental point

$$\xi_1 : \xi_2 : \xi_0 = 0 : 1 : 0.$$

§ 13. Examples of relatively integral domains of the first kind and of regular systems.

1. As a first example of a relatively integral domain of the first kind $\mathfrak{S}_{nn}$, we consider the *totality of the polynomials in $x_1, \dots, x_n$ that admit a given permutation group G* — thus the totality of the polynomials of a Lagrange genus domain. Since $x_1, \dots, x_n$, by virtue of the identity

$$x_k^n - \sigma_1(x)x_k^{n-1} + \sigma_2(x)x_k^{n-2} + \dots \pm \sigma_n(x) = 0 \quad (k = 1, 2, \dots, n), \quad (1)$$

are algebraically integral over the elementary symmetric functions belonging to $\mathfrak{S}_{nn}$, and hence by Definition V are algebraically integral over $\mathfrak{S}_{nn}$, $\mathfrak{S}_{nn}$ is of the first kind. Moreover $\mathfrak{S}_{nn}$, as a domain of the first kind, since its elements are homogeneous forms and $n = \rho$, forms a regular system (by the conclusion of § 10). Indeed, because of (1), the resultant R_0 of the elementary symmetric functions cannot vanish identically; by the rules of calculation of resultant theory, R_0 is computed to be ± 1 . The *involution form* of $\mathfrak{S}_{nn}$ is uniquely determined — since $\mathfrak{S}_{nn}$ is of the first kind and $n = \rho$ — as the involution form of the corresponding Lagrange genus domain. Since the quantities conjugate to $x_1, \dots, x_n$ with respect to this domain arise through the permutations of the group G , this involution form is therefore given by

$$\begin{aligned} \Psi(z, u) &= \prod_i (z - u_1 x_{i_1} - u_2 x_{i_2} - \dots - u_n x_{i_n}) \\ &= z^i + \sum' G_{\alpha\alpha_1 \dots \alpha_n}(x) z^\alpha u_1^{\alpha_1} \dots u_n^{\alpha_n}, \end{aligned} \quad (2)$$

where the product is to be extended over all permutations of G . Formula (2) represents the “Galois form of the group G ,” and hence the following fact holds:

All polynomials in $x_1, \dots, x_n$ which admit a permutation group G are integral rational combinations of the coefficients $G_{\alpha\alpha_1 \dots \alpha_n}(x)$ of the Galois form of the group.

2. As an example of *regular systems*, consider the systems $\mathfrak{S}_{n1}$ of polynomials, that is, systems which contain only one algebraically independent polynomial.

Indeed, let $\mathfrak{S}_{11}^*$ be a mapping system of $\mathfrak{S}_{n1}$ and let

$$F^*(x) = a_0 x^k + a_1 x^{k-1} + \dots + a_k \quad (a_0 \neq 0)$$

be a polynomial from $\mathfrak{S}_{11}^*$. Then

$$R_0 = a_0 \neq 0. \quad (3)$$

Thus $\mathfrak{S}_{n1}$ is regular by Definition VIII, and consequently:

*Every system $\mathfrak{S}_{n1}$ of polynomials, in particular every system of infinitely many polynomials in one indeterminate, possesses an integrality basis.*¹⁵

3. We now specialize the systems $\mathfrak{S}_{n1}$ to relatively integral domains $\mathfrak{S}_{n1}$, which, as regular systems by § 12, are relatively integral domains of the first kind. Their integrality basis is therefore given by an involution basis, which is at the same time the involution basis of the smallest containing field $\mathfrak{K}_{n1}$. But by § 6, any function of the involution basis of $\mathfrak{K}_{n1}$ is at the same time a minimal basis — we called it the Lüroth function of the field — and every function of the involution basis depends on this Lüroth function by a fractional-linear transformation. In the present case the Lüroth function, since it belongs to $\mathfrak{S}_{n1}$, is a polynomial; every further polynomial of the involution basis therefore depends fractional-linearly, and by (3) algebraically integrally, hence integrally and linearly, on this Lüroth function $L(x)$, which thus becomes the integrality basis. The involution form of $\mathfrak{S}_{11}^*$, setting also ($u_1 = 1$), is

$$\Psi^*(z) = L^*(z) - L^*(x),$$

from which $L(x)$ again appears as integrality basis. Thus:

The relatively integral domains $\mathfrak{S}_{n1}$ are of the first kind; their integrality basis consists of one polynomial, the Lüroth function of the smallest containing field.

4. As an example of a relatively integral domain $\mathfrak{S}_{n1}$, let us mention specifically the integral rational projective invariants of a quadratic form in s variables. In agreement with the preceding, it is well known that they can be represented as integral rational combinations of the determinant $|a_{ik}|$ of the form; and this determinant is at the same time the Lüroth function of the corresponding invariant field.

¹⁵The simplest possible non-regular systems without integrality basis are therefore systems $\mathfrak{S}_{22}$. That such systems actually exist is shown by Hilbert's example and by the example given in the note to § 12.

§ 14. Systems consisting of integer polynomials.

The basis theorems obtained are now to be sharpened with respect to integrality over the integers.

The coefficient domain Ω will therefore in what follows be assumed to be a (finite) algebraic number field; and let $[\Omega]$ denote the totality of the algebraic integers from Ω , which we shall briefly call whole numbers. By an integer polynomial is meant a polynomial with coefficients from $[\Omega]$; from now on $F(x), G(x), \dots$ shall denote only integer polynomials.

In analogy with § 7, we consider the different systems made up of integer polynomials:

An arbitrary system of integer polynomials shall be denoted as an integer system $\mathfrak{S}_{n\rho}$.

The integer linear family $\mathfrak{L}_{n\rho}$ is an integer system which, along with $F_1(x)$ and $F_2(x)$, always contains $c_1 F_1(x) + c_2 F_2(x)$, where c_1, c_2 are arbitrary whole numbers from $[\Omega]$.

The integer integrality domain $\mathfrak{I}_{n\rho}$ is an integer linear family which, along with $F(x)$ and $G(x)$, always contains $F(x) \cdot G(x)$.

The integer relatively integral domain $\mathfrak{G}_{n\rho}$ is an integrality domain which, along with $F(x)$ and $G(x)$ — where $G(x) \neq 0$ — always and only contains the quotient $F(x) : G(x)$ when this quotient is an integer polynomial.

Correspondingly as in § 7, the smallest containing integer integrality domain $\mathfrak{I}_{n\rho}$ of an integer system $\mathfrak{S}_{n\rho}$ consists of all polynomials $H(x)$ which are integral rational, integer combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$;

the smallest containing integer relatively integral domain $\mathfrak{G}_{n\rho}$ consists of all integer polynomials $H(x)$ which are rational combinations of a finite number of polynomials from $\mathfrak{S}_{n\rho}$.

We now pass to the transfer of the basis theorems. With respect to the rational basis, it is to be noted that the notions of field, smallest containing field, and rational basis, which rest only on “rationality,” undergo no change. Furthermore, the mapping system $\mathfrak{S}_{\rho\rho}^*$ of an integer system $\mathfrak{S}_{n\rho}$ can in arbitrarily many ways again be chosen as an integer system. Since, also for the integer linear family, the arguments leading to the bound on the number of generators remain valid, one obtains:

Theorem V': Every integer system $\mathfrak{S}_{n\rho}$ possesses a rational basis; the rational basis of an integer linear family $\mathfrak{L}_{n\rho}$ can in particular be chosen so that it contains at most $\rho + 1$ polynomials.

We now investigate the analogue of the involution basis of integrality domains of polynomials (§ 8). For this purpose the theorem on *symmetric functions of rows of quantities* must be extended with respect to *integrality over the integers* by the following

Auxiliary theorem. The integral integer symmetric functions of n rows of quantities $(yz \cdots t)$:

$$\begin{array}{cccc} y_1 & z_1 & \cdots & t_1 \\ y_2 & z_2 & \cdots & t_2 \\ \vdots & \vdots & & \vdots \\ y_n & z_n & \cdots & t_n \end{array} \quad (1)$$

are integral integer functions of a finite number among them, namely of the elementary symmetric functions

$$\begin{array}{c} \sigma_1(y), \sigma_2(y), \dots, \sigma_n(y); \quad \sigma_1(z), \sigma_2(z), \dots, \sigma_n(z); \quad \dots; \\ \sigma_1(t), \dots, \sigma_n(t); \end{array} \quad (2)$$

and of the functions

$$\chi_1(y, z, \dots, t), \dots, \chi_k(y, z, \dots, t), \quad (3)$$

which are algebraically integral and integer over the functions (2).

Indeed, the n quantities $y_1, \dots, y_n$ depend algebraically integrally and integrally over the integers on $\sigma_1(y), \dots, \sigma_n(y)$; the corresponding assertion holds for $z_1, \dots, z_n$ with respect to $\sigma_1(z), \dots, \sigma_n(z)$, and so on. The field of symmetric functions of the rows of quantities (1) therefore forms an algebraic field over the field determined by the functions (2), in such a way that the algebraically integral and integer quantities of this field are identical with the integral integer symmetric functions of the rows of quantities (1). If one therefore takes (3) as Kronecker's fundamental system, the auxiliary theorem is proved.

Let now $\mathfrak{I}_{n\rho}$ be an integer integrality domain, $\mathfrak{I}_{\rho\rho}^*$ an integer mapping domain, and

$$\Psi^*(z, u) = G^*(x)z^i + \sum' G_{\alpha\alpha_1 \dots \alpha_\rho}^*(x) z^\alpha u_1^{\alpha_1} \cdots u_\rho^{\alpha_\rho}$$

an involution form of $\mathfrak{J}_{\rho\rho}^*$; here $G^*(x)$ may also have dimension zero, that is, be a number from $[\Omega]$. The elementary symmetric functions corresponding to the functions (2) are then given by certain ones among the quotients

$$G_{\alpha_1 \dots \alpha_\rho}^*(x) : G^*(x). \quad (4)$$

Since by the auxiliary theorem the functions (3) depend algebraically integrally and integrally over the integers on (2), the functions from $\mathfrak{K}_{\rho\rho}^*$ corresponding to (3) are given — by the extension of Gauss's theorem to polynomials with algebraically integral coefficients — by

$$K_\nu^*(x) : (G^*(x))^\sigma, \quad (5)$$

where $K_\nu^*(x)$ are integer polynomials from $\mathfrak{J}_{\rho\rho}^*$. Thus $K_\nu^*(x)$ belong to the domain if the case is that of relatively integral domains; in general they are quotients of polynomials from $\mathfrak{J}_{\rho\rho}^*$.

If now $F^*(x)$ is an arbitrary integer polynomial from $\mathfrak{J}_{\rho\rho}^*$, then one has

$$F^*(x) = \frac{1}{i} \left(F^*(x) + F^*(x^{(1)}) + \dots + F^*(x^{(i-1)}) \right),$$

and therefore $i \cdot F^*(x)$ becomes an integral integer symmetric function of the rows of quantities $x, x^{(1)}, \dots, x^{(i-1)}$.

By the auxiliary theorem and the transfer principle, therefore:

Theorem VI': For integer integrality domains $\mathfrak{J}_{n\rho}$, each involution form determines a distinguished rational basis in such a way that every polynomial $F(x)$ from $\mathfrak{J}_{n\rho}$ admits a representation

$$F(x) = \frac{\Gamma(H(x), H_1(x), \dots, H_s(x))}{i \cdot H(x)^i},$$

where Γ denotes an integral integer function. For integer relatively integral domains $\mathfrak{S}_{n\rho}$, with integer relatively integral domain $\mathfrak{S}_{\rho\rho}^*$ as mapping domain, the denominator $H(x)$ is given by the highest coefficient of the corresponding involution form.

§ 15. Integer relatively integral domains of the first kind and regular systems.

For integer systems, the integer integrality basis takes the place of the integrality basis; that is, a finite number of integer polynomials of the system such that every integer polynomial of the system can be represented as an integral integer combination of this finite number.

The theorems on the integer integrality basis run essentially parallel to the earlier ones, and only the proofs require slight modifications.

First, Definition V (§ 9) transfers directly.

Definition V': A quantity ξ is called algebraically integral and integer with respect to the integer relatively integral domain $\mathfrak{S}_{n\rho}$ if it satisfies some equation whose highest coefficient is a unit from $[\Omega]$ and whose remaining polynomials are from $\mathfrak{S}_{n\rho}$.

The extension of Gauss's theorem to polynomials with algebraically integral coefficients shows, as in § 9, that the definition at the irreducible equation can be obtained from this.

From Definition V' one further obtains the transfer of Definition VII.

Definition VII': An integer relatively integral domain $\mathfrak{S}_{n\rho}$ is called of the first kind if, as mapping domain, it possesses a relatively integral domain $\mathfrak{S}_{\rho\rho}^$ such that every arbitrary integer polynomial in $x_1, \dots, x_\rho$ is algebraically integral and integer over $\mathfrak{S}_{\rho\rho}^*$.*

To show the finiteness of these domains of the first kind, we first note that, as in § 10, it follows from the definition that the highest coefficient of the involution form is a unit from $[\Omega]$.

By Theorem VI' (§ 14), every polynomial $F(x)$ from $\mathfrak{S}_{n\rho}$ therefore admits a representation

$$F(x) = \frac{\Gamma(H_1(x), \dots, H_\sigma(x))}{i}.$$

Applying here to the system formed from the integer polynomials $\Gamma(H_1, \dots, H_\sigma)$ Hilbert's Theorem II on the integer module basis (Math. Ann. 36), originally stated for integral rational numerical coefficients, in its

extension to algebraically integral polynomials,¹⁶ the existence of the integer integrality basis follows; that is:

Theorem VII': Every integer relatively integral domain of the first kind possesses an integer integrality basis.

We further transfer the definition of regular systems:

Definition VIII': An integer system $\mathfrak{S}_{n\rho}$ is called integer-regular if, in a mapping domain $\mathfrak{J}_{\rho\rho}^$ of the smallest containing integer integrality domain, there exist ρ polynomials such that the resultant of the terms of highest dimension is a unit from $[\Omega]$ — $R_0 = \varepsilon$.*

In order to transfer the existence proof of the integrality basis, we note that the auxiliary theorem of § 11 admits the following sharper formulation, which follows from equations (5) and (6) in § 11 and from the fact that $A_1(\lambda), \dots, A_r(\lambda)$ are integral integer functions of $\lambda_1, \dots, \lambda_\rho$:

A sufficient condition that every arbitrary integer polynomial in $x_1, \dots, x_\rho$ depend algebraically integrally and integrally over ρ integer polynomials is that the resultant of the terms of highest dimension of these polynomials be a unit from $[\Omega]$ — $R_0 = \varepsilon$.

From this auxiliary theorem, exactly as in § 12, for every polynomial $F(x)$ of an integer-regular system, representation (3) of § 12 follows, where now $\Gamma_1(G_i), \dots, \Gamma_k(G_i)$ are integer polynomials in the G_i . Furthermore, as in § 12, the application of Hilbert's Theorem II (Math. Ann. 36) in its extension to algebraically integral polynomials, together with the definition of the smallest containing integer integrality domain, gives the existence of the integer integrality basis; that is:

Theorem IX': Every integer-regular system possesses an integer integrality basis.

As an example of an integer relatively integral domain of the first kind, in analogy with example 1 in § 13, we refer to the totality $\mathfrak{S}_{nn}$ of integer polynomials of a Lagrange genus domain. That $\mathfrak{S}_{nn}$ is integer of the first kind follows from the identity

$$x_k^n - \sigma_1(x)x_k^{n-1} + \dots \pm \sigma_n(x) = 0 \quad (k = 1, 2, \dots, n);$$

since the resultant of the elementary symmetric functions has the value ± 1 , $\mathfrak{S}_{nn}$ is also an integer regular system.

A further example is given, by the auxiliary theorem of § 14, by the totality of the integral integer symmetric functions of n rows of quantities.

Erlangen, May 1914.

¹⁶If $w_1, \dots, w_k$ denotes a fundamental system of the algebraic integers from Ω , and one sets

$$\Gamma(H_1, \dots, H_\sigma) = \Gamma_1(H)w_1 + \dots + \Gamma_k(H)w_k,$$

then $\Gamma_1(H), \dots, \Gamma_k(H)$ have integral rational numerical coefficients. Applying Theorem II to the system formed from the forms $v_1\Gamma_1(H) + \dots + v_k\Gamma_k(H)$ directly shows the validity of the extension.